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The loop invariant for NFS

Q = the post-condition is
"All vertices have ~~been~~ visited"
(marked)

P = the precondition.
= everything we know to be true before
entering the loop the first time

G = the loop guard

Q = post-condition, what we hope the loop will do.

We want to prove:

- like a proof by induction
- ① INITIALIZATION $P \Rightarrow L$
"Upon first reaching the loop, L is true"
 - ② MAINTENANCE $L_i \wedge G \Rightarrow L_{i+1}$
 L_i = the loop inv. going into the i th iteration
of the loop.
"If I'm able to enter the loop & the loop inv. was true,
then when I try to re-enter next time,
the loop inv. will be true again"
 - ③ END

$$L \wedge \neg G \Rightarrow Q$$

"If the loop inv. true & if I exit the
loop, then I did what I was supposed to do"

Correctness is two steps:

① Partial Correctness
(usu. via a loop invariant)

"If the alg terminates,
then it is correct"

② Termination

"The algorithm terminates"

$\therefore$ The algorithm is
correct.

$$\boxed{\begin{array}{c} A \Rightarrow B \\ A \\ \therefore B \end{array}}$$

The loop invariant is: [will add to it as needed]

(a) $\bar{X}$ is a set of unmarked vertices in V such that $\forall x \in \bar{X}, \exists$ a path from v_0 to x such that all vertices on the path are marked (except x itself).

(b) Our graph $G = (V, E)$ is ^{path} connected

(c) If $v_0 \notin \bar{X}$, then v_0 is marked.

(d) For all marked vertices v , if w is a neighbor of v , then either $w \in \bar{X}$ or w is marked.

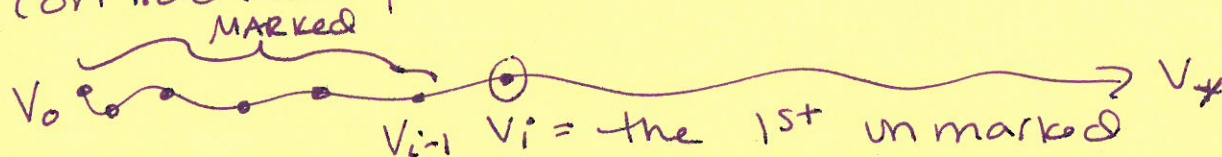
Proof of END: $L \wedge \neg G \Rightarrow Q$

Let ^{(c) + (d)} (a) and (b) are true. ~~and~~

Let $\neg G$ be true, then, $\bar{X} = \emptyset$.

Let $v_* \in V$. By (b), we know that there exists a path from v_0 to v_* .

By contradiction, assume v is not marked.



By (c), we know $v_i \neq v_0 \Rightarrow i \geq 1 \Rightarrow v_i$ has a left neighbor.
[Note, the vertex v_i is connected to the previous vertex v_{i-1} , which is marked.]

In (d), let $v = v_{i-1}$ and $w = v_i$ & we have a contradiction! ~~XXXX~~